

JAYLANI GAILANI

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EDUCATION

Rutgers University — B.S., Electrical Engineering

New Brunswick, NJ

Expected December 2027

- **Achievements:** Dean's List | Cullen Family Endowed Scholarship | Constellation Engineering Scholarship | James & Edna Noe Endowed Scholarship | Harold Borkan Endowed Scholarship
- **Relevant Coursework:** Principles of Electrical Engineering I & II, Digital Logic Design, Computer Architecture & Assembly Language, Linear Systems & Signals, Electronic Devices, Differential Equations, Engineering Mechanics – Statics, Probability Theory

RESEARCH EXPERIENCE

Undergraduate Research – Interactive Computer Architecture Web Platform

Rutgers University, Department of Electrical Engineering

- Contributed to an educational web platform researching and visually teaching core computer architecture concepts to undergraduate students, including instruction execution, system calls, and CPU security vulnerabilities
- Rebuilt the platform's x86 assembler module into a functioning instruction-level assembler and built an animated visualization of system call handling using signals
- Tracked and delivered development tickets through Linear and used Git/GitHub for version control across a multi-contributor codebase

PROJECTS

Face-Tracking Camera Turret *Python, OpenCV, Arduino*

- Designed and built a 2-axis (pan-tilt) computer-vision turret that detects and tracks a human face in real time, splitting duties between a PC (OpenCV face detection) and an Arduino Uno (motor control) over USB serial
- Engineered a custom 3D-printed pan-tilt mount and isolated the servo power supply from the microcontroller to prevent brownouts and current spikes under load
- Implemented proportional angle control with dead-zone logic and slew-rate limiting to eliminate jitter and overshoot from motor inertia

Dementia Assist — Object Location Memory System *ESP32-CAM, Python, YOLOv8*

- Developed an offline object-tracking pipeline using YOLOv8-nano to detect and log the last known location of everyday items from an ESP32-CAM video stream.
- Designed a multithreaded MJPEG stream reader to eliminate detection lag and built an offline text-to-speech query interface to retrieve saved locations.

12-Bus Transmission & Distribution Reliability Analysis *PowerWorld Simulator*

- Simulated a 230 kV / 115 kV transmission network under a 15% peak demand surge, identifying and resolving thermal overloads up to 111% on constrained sub-transmission lines.
- Verified N-1 contingency compliance by tripping the primary 230 kV backbone tie-line and confirming power rerouted through secondary paths with maximum loading of 95%.

Custom ESP32 Sensor Hub PCB *KiCad*

- Designed and routed a 2-layer PCB integrating an ESP32-S3, environmental and motion sensors, and regulated 3.3V power distribution.
- Implemented a continuous ground plane with strategic via transitions and passed KiCad's Design Rules Checker with zero errors prior to fabrication at JLCPCB.

TECHNICAL SKILLS

Hardware & Lab: Oscilloscope, Multimeter, Signal Generator, Power Supply, Breadboard Prototyping, Soldering, 3D Printing
Microcontrollers & Interfaces: Arduino (ATmega328P), ESP32/ESP32-CAM, I2C, SPI, UART, ADC/DAC, Serial Communication, Closed-Loop Control

Digital Logic: Finite State Machines, TTL Logic, Flip-Flops, 555-Timer Circuits, Closed-Loop Feedback Control

Programming & Tools: Python (OpenCV, NumPy, Matplotlib), C/C++, MATLAB, Java, HTML/CSS/JavaScript, Git/GitHub, Linear

PCB Design: KiCad, Schematic Capture, 2-Layer PCB Layout & Routing, DRC

Power Systems: PowerWorld Simulator, Load Flow Analysis, Short-Circuit/Fault Analysis, N-1 Contingency Analysis, Transformer Sizing & Tap Control, Reactive Power Compensation